# MoCafe v2.00 — A Monte-Carlo Dust RT Code with Scattering and Thermal Emission: User Guide

**Abstract**

MoCafe is a parallel (Fortran 90 + MPI) Monte-Carlo radiative-transfer code for dust scattering and thermal emission. It transports photon packets over a wavelength grid through a dusty medium, scatters them off dust grains (Henyey–Greenstein phase function or a tabulated Mueller matrix), tallies the radiation field in every cell, and re-emits the absorbed energy as thermal dust emission through either the Lucy (1999) method coupled to the SEDUST grain library (equilibrium and stochastically heated grains, including PAH features) or the Bjorkman & Wood (2001) immediate-reemission method. It supports multiple stellar populations and galaxy models, an all-sky interior observer (for the Milky-Way case), and a Modified Random Walk for very optically thick regions. This guide covers installation, execution, the complete input reference, the dust-emission workflow, the output products, worked examples, and the underlying algorithms.

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# 1. Overview

MoCafe follows photon packets emitted by one or more sources through a dusty medium. Packets scatter off dust grains (Henyey–Greenstein phase function or a tabulated Mueller matrix) and are absorbed and re-emitted as thermal radiation. At every interaction a contribution is "peeled off" toward each observer to build an image or spectral-energy distribution (SED).

**Capabilities.**

- **Multi-wavelength (SED) transport** (`par%use_sed`): one run covers the full UV-to-far-IR range on a log-spaced wavelength grid.
- **Mean intensity in each cell** $J_\lambda(x,y,z)$ (`par%save_jlam`) via the Lucy (1999) pathlength estimator.
- **Dust thermal emission** (`par%use_dustemis`) in two modes: *Mode 1* (Lucy + SEDUST, grain models astrodust / DL07 / Zubko) and *Mode 2* (Bjorkman & Wood 2001).
- **Galaxy models**: multiple stellar populations (`par%nsource`) with radially-exponential and sech$^2$ disks, log-spiral arms, and Sersic (alias-sampled) / boxy / bar bulges.
- **All-sky interior observer** (`par%allsky`) for the Milky-Way case.
- **Modified Random Walk** (`par%use_mrw`) plus Russian roulette for very optically thick media.
- **Python analysis tools** (`python/`): a format-agnostic reader and plotters for SEDs, temperature/power maps, $J_\lambda$, and all-sky maps.

The emission features are opt-in: a plain monochromatic scattering run uses none of them. The design and validation record is in `MoCafe_v2.00_PLAN.md`.

**Units.** In scattering-only mode MoCafe is unit-agnostic and the image is in `(luminosity unit)` $\mathrm{cm}^{-2}\,\mathrm{sr}^{-1}$. In SED / emission mode a physical length scale is required (through `par%distance_unit`: one grid length unit equals one distance unit) and the source luminosity (`par%luminosity`) is in $\mathrm{erg\,s}^{-1}$, so intensities and dust temperatures are absolute.

# 2. Installation

## 2.1 Prerequisites

- A Fortran 2008 MPI compiler: Intel oneAPI (`mpiifort` or `mpiifx`) or GNU (`mpif90`). MoCafe is MPI-only.
- **CFITSIO** (required).
- **HDF5** 1.14+ with Fortran bindings built by the same compiler (optional; on by default).
- Python 3 with `numpy`, `astropy`, `h5py` (and `scipy` for the Illustris converter) for the analysis tools.

## 2.2 Building MoCafe

```
make                              # HDF5=1 (default) -> MoCafe.x
make HDF5=0                       # CFITSIO only
make HDF5_PREFIX=/usr/local/hdf5  # override HDF5 location
make F90=mpif90                   # force a compiler
make DEBUG=1                      # checked / traceback flags
```

The compiler is auto-selected `mpiifort` → `mpiifx` → `mpif90`. The build always passes `-cpp -DMPI` and (by default) `-DHDF5`. When you add a source file, append its object to `OBJS` in the `Makefile` in dependency order.

## 2.3 Building the SEDUST library

The Lucy emission engine (Mode 1) links the SEDUST library, kept self-contained under `SEDust/`. Build it once:

```
cd SEDust/sed && ./build_lib.sh    # -> SEDust/sed/lib/libsedust.a  (Intel)
```

The MoCafe `Makefile` links `SEDust/sed/lib/libsedust.a` automatically (variable `SEDUST_LIBDIR`). The SEDUST module named `mathlib` is renamed to `sed_mathlib` to avoid a clash with MoCafe's own `mathlib`. Each model's whole optics is one HDF5 product, `SEDust/data/<model>/sedust_<model>.h5` (∼44 MB for the six), and those are committed to the repository, and `par%sed_datadir` defaults to blank so MoCafe auto-resolves the SEDUST data directory relative to the `MoCafe.x` executable at run time — a fresh checkout runs dust emission from any working directory with no path editing. That directory is the data root for the whole build, the dielectric functions included, so there is no directory to change into either. What is kept out of git is what no run with the products opens: the 17 MB D16 spheroid Q-library (used by a non-default graphite path), the text Q tables and extinction curves, and the 25 MB distributed ZDA optics tables, which `sedust_zubko.h5` carries as its default `/qtable` and which only an `HDF5=0` build needs. Run `SEDust/populate_data.sh` to fetch them, editing its source path (or setting `SEDUST_SRC`) to your own SEDUST tree. Mode 2 (Bjorkman & Wood) needs only the extinction table and not SEDUST.

# 3. Running MoCafe

```
mpirun -np <N> ./MoCafe.x <input>.in
```

The single argument is a Fortran namelist file. Every tunable quantity is a field of the global `par` structure and is set as `par%<name> = <value>` inside a `&parameters / . . . /` block. Sample inputs and `run.sh` scripts live under `examples/`. The default of every parameter is the initializer of `params_type` in `src/define.f90`.

**Reproducibility.** A run is bit-identical when `par%iseed` is a fixed non-zero value, `par%use_master_slave` is `.false.` (equal-share), and the number of MPI ranks is fixed (`par%iseed` = 0 seeds from the clock/PID). With `par%launch_sequence ='sobol'` the launch set is indexed by the global photon number and is therefore independent of the MPI task count: the internal-source `Direct0` image is bit-identical for any `-np`, while the scattered field still

follows the rank-local transport streams.

# 4. Input Parameter Reference

Parameters are grouped below. Types: `L` logical, `I` integer, `R` real, `S` string, `[ ]` array.

## 4.1 Simulation control

| Parameter | Default | Description |
|---|---|---|
| `no_photons` | 1e5 | Number of photon packets |
| `no_print` | 1e7 | Progress-print interval (photons) |
| `iseed` | 0 | RNG seed (0 → seeded from clock/PID) |
| `launch_sequence` | 'random' | 'sobol' = quasi-random (Owen-scrambled Sobol) launch of every packet; transport draws stay pseudo-random (below) |
| `qmc_seed` | 12345 | Scramble seed for 'sobol'; a different seed is an independent replicate |
| `use_master_slave` | .true. | MPI mode: master–slave vs. equal-share |
| `num_send_at_once` | 10000 | Batch size (master–slave) |
| `luminosity` | 1.0 | Total source luminosity [erg/s] (SED/emission mode) |

**Quasi-random photon launching.** With `par%launch_sequence ='sobol'` the *launch* variables of each packet — the compose origin, the source component, the wavelength (inverted through the monotonic luminosity CDF), the emission direction, and the external-sphere entry point — are taken from an Owen-scrambled Sobol point indexed by the global photon number, while every draw after launch (forced first scattering, Henyey–Greenstein, free paths) stays on the Mersenne Twister; the transport estimator is unchanged and unbiased. The packet count in a wavelength bin of probability $p$ then deviates from $pN$ by order one (instead of binomially), the `Direct`/`Direct0` images lose their launch noise, and the launch set is independent of the MPI task count (the `Direct0` cube is bit-identical for any `-np`). `par%qmc_seed` selects the scramble; several seeds give independent replicates for error estimation. All three external boundaries (`sph`, `rec`, `cyl`) have quasi-random entry mappings, and the dust-emission packets use a second, decorrelated scramble stream, reused across the Lucy iterations as common random numbers. Not combined (setup error) with `par%use_stokes`, the $(a, g)/\tau$ scans, or `par%radiation_angular_PDF_file`. The helper `python/qmc_replicates.py` runs a namelist over several scrambles and reports the replicate RMS and the effective photon-number gain. Design and validation: `docs/QUASI_RANDOM_LAUNCH_MOCAFE.md`.

## 4.2 Grid geometry (Cartesian)

| Parameter | Default | Description |
|---|---|---|
| `nx, ny, nz` | 1,1,11 | Grid resolution |

| Parameter | Default | Description |
|---|---|---|
| `xmax, ymax, zmax` | 1.0 | Half-sizes of the box (grid spans $[-\max, \max]$) |
| `rmax` | -999 | If $> 0$ and not a cylinder, forces a cube of side $r_{\max}$ (sphere) |
| `nr` | -999 | If $> 1$, sets nx=ny=nz=nr |
| `geometry` | ' ' | '', 'sphere', 'cylinder', or 'Plummer' (auto-set from source) |
| `xyz_symmetry` | .false. | Exploit octant symmetry (disabled if any $n = 1$) |
| `z_symmetry` | .false. | Exploit reflection about $z = 0$ |
| `xy_periodic` | .false. | Periodic in $x, y$ |

## 4.3 Grid type and media

| Parameter | Default | Description |
|---|---|---|
| `grid_type` | 'car' | 'car' (Cartesian), 'clump' (clumpy), 'amr' (octree) |
| `amr_file` | ' ' | Generic AMR octree file (FITS/HDF5/text) for 'amr' |
| `dust_density_law` | 'global_dgr' | AMR leaf opacity: 'global_dgr', 'from_file', 'laursen09' |
| `DGR, Z_global, Z_ref, f_ion_dust` | — | Dust-to-gas / metallicity parameters (AMR) |
| `clump_radius` | -1 | Clump radius (clumpy medium) |
| `clump_f_cov / _f_vol / _N_clumps` | -1 | Clump count specification |
| `clump_tau0 / _ndust / _nH` | -1 | Opacity specification for each clump |

The clumpy and AMR media are documented in `docs/MoCafe_clump.pdf` and `docs/MoCafe_amr.pdf`. The full dust-emission suite runs on both the Cartesian grid and the AMR octree (§5.8); the clumpy medium is scattering-only.

## 4.4 Distance unit

| Parameter | Default | Description |
|---|---|---|
| `distance_unit` | 'kpc' | '', 'pc', 'au', 'kpc'; sets `distance2cm`. In SED mode one grid unit = one distance unit (physical scale) |

## 4.5 Source geometry (single source)

| Parameter | Default | Description |
|---|---|---|
| `source_geometry` | 'point' | 'point', 'uniform', 'uniform_xy', 'gaussian', 'exponential', 'sech', 'exp_spiral', 'external_{sph,cyl,rec}' |

| Parameter | Default | Description |
|---|---|---|
| `xs_point, ys_point, zs_point` | 0.0 | Point-source location |
| `source_zscale` | NaN | Vertical scale height (gaussian/exponential/sech/exp_spiral) |
| `source_rscale` | NaN | Radial scale length of the exponential disk |
| `radiation_angular_PDF_file` | '' | Angular PDF for external illumination |
| `ext_geometry` | '' | External-field boundary *when composing* an internal source with an external field: 'sph'/'cyl'/'rec' (blank auto-selects from geometry/rmax); the external-only run uses source_geometry='external_*' instead (§5.1) |

Multiple sources are described in §5.5.

## 4.6 Source configurations: monochromatic vs SED

MoCafe runs one or more internal sources, an external isotropic radiation field, or an internal source composed with an external field, in either the monochromatic (scattering-only) mode or the SED mode (Table 6). Multiple internal sources (`par%nsource` $> 1$, §5.5) work in *both* modes, not only in SED; a single internal source uses the scalar `par%source_geometry`. The external-only field uses `source_geometry='external_{sph,cyl,rec}'`, while the composed internal + external case (§5.1) is switched on by adding an external field to a non-external internal source.

Table 6: Supported source configurations: monochromatic scattering vs. SED, and single / multiple internal / external / composed sources.

| Source configuration | Monochromatic (scattering only) | SED (spectrum → dust emission) |
|---|---|---|
| Single internal source (point / uniform / gaussian / disk / bulge) | supported | supported |
| Multiple internal sources (`par%nsource` $> 1$) | supported | supported |
| External isotropic field (`source_geometry='external_*'`) | supported | supported |
| Internal source(s) + external field composed in one run | supported | supported |

## 4.7 Dust and scattering (grey / monochromatic)

| Parameter | Default | Description |
|---|---|---|
| `albedo` | 0.3253 | Single-scattering albedo |
| `hgg` | 0.6761 | Henyey–Greenstein asymmetry $g$ |

| Parameter | Default | Description |
|---|---|---|
| `cext_dust` | 1.61e-21 | Dust extinction per H nucleus [$cm^2$] |
| `scatt_mat_file` | ' ' | Mueller-matrix table (`data/mueller_*.dat`); enables Stokes |
| `use_stokes` | .true. | Stokes polarization (auto-off without a matrix file) |

In SED mode the wavelength-dependent albedo, $g$, and $C_{\rm ext}$ come from the `par%dust_model` grain model and override these scalars (§5.1).

## 4.8 Optical depth and density

| Parameter | Default | Description |
|---|---|---|
| `taumax` | -999 | Radial optical depth, center → edge (sets the density scale) |
| `tauhomo` | -999 | Optical depth of the volume-equivalent homogeneous medium |
| `density_file` | ' ' | 3-D density cube (FITS/HDF5) |
| `reduce_factor` | 1 | Density-file down-sampling |
| `centering` | 0 | Density-file re-centering mode |
| `density_zscale`, `density_rscale`, `density_powerlaw_index` | NaN | Analytic density modulations |

In SED mode `par%taumax`/`par%tauhomo` are defined at the reference wavelength `par%lambda_ref`.

## 4.9 Peel-off observers

| Parameter | Default | Description |
|---|---|---|
| `nxim, nyim` | 129 | Image pixels |
| `dxim, dyim` | NaN | Pixel size [deg] (auto to cover the grid if unset) |
| `distance` | NaN | Observer distance; renormalizes (obsx,obsy,obsz) |
| `inclination_angle`, `position_angle, phase_angle` | NaN | Observer angles [deg] (arrays) |
| `alpha, beta, gamma` | NaN | Euler angles [deg] (alternative) |
| `obsx, obsy, obsz` | NaN | Explicit observer direction (alternative) |

Up to `MAX_OBSERVERS = 181` observers per run. Angle conventions are in `docs/MoCafe_Geometry.pdf`.

## 4.10 Output control

| Parameter | Default | Description |
|---|---|---|
| `file_format` | 'hdf5' | 'hdf5' or 'fits'; authoritative for the container |
| `out_file` | ' ' | Output base name (from the input name if empty) |
| `out_bitpix` | -32 | FITS bitpix (−32 float32, −64 float64) |
| `save_direc0` | .false. | Also write the unattenuated direct image |
| `sightline_tau` | .false. | Also write a sight-line optical-depth map for each pixel |

# 5. Dust Thermal Emission

The emission workflow is: *SED transport* (panchromatic scattering) → $J_\lambda$ *tally* → *dust emission* → observer SEDs. Enable it with `par%use_sed`, and add `par%save_jlam` and `par%use_dustemis` for the emission.

## 5.1 Multi-wavelength (SED) mode

| Parameter | Default | Description |
|---|---|---|
| `use_sed` | .false. | Enable panchromatic (multi-wavelength) scattering transport |
| `nlambda` | 128 | Number of log-spaced wavelength bins (binning only; wavelengths are continuous) |
| `lambda_min, lambda_max` | 0.0912, 2000 | Wavelength range [$\mu$m] |
| `lambda_ref` | 0.55 | Reference wavelength [$\mu$m] for the grid opacity and $\tau$ |
| `kext_file` | ' ' | *Override:* explicit table $\lambda$, $\varpi$, $\langle\cos\rangle$, $C_{\rm ext}$/H; blank = take them from `par%dust_model` |
| `source_spectrum` | ' ' | Two-column source spectrum $\lambda$, $L_\lambda$ (columns set by spectrum_type) |
| `spectrum_type` | 'shape' | Column units of every source spectrum file (below) |
| `tstar` | -999 | Planck source temperature [K] (if no spectrum file) |
| `save_jlam` | .false. | Tally $J_\lambda$ in each cell; *opt-in*, required by the `'lucy'` dust emission (§5.) |
| `use_dustemis` | .false. | Add dust thermal emission (§5.); **leave both this and save_jlam off for scattering only** over the wavelength range |

A packet's wavelength is drawn *continuously* from the source spectrum, by inverting its cumulative luminosity; `par%nlambda` sets how the results are binned — the output image planes, the $J_\lambda$ tally, the dust-emission spectra — and not how finely the spectrum is sampled. Every wavelength-dependent quantity a packet carries ($s_{\rm ext}$, albedo, $g$) is read at its own wavelength, so a coarse `par%nlambda` costs output resolution but does not bias the transport. The same holds for the wavelengths of reemitted dust packets.

Those quantities are the size-integrated optics of the grain model named by `par%dust_model`, evaluated by SEDUST

from the same model object that supplies the dust emission. Naming the model is therefore enough to fix the dust physics of a run: the energy a cell absorbs and the spectrum it radiates back come from one size distribution and one set of cross sections. Every model carries its scattering optics — `astrodust` from its T-matrix $Q$ table, `dl07` and `mrn` from Mie theory on the astrosilicate and graphite dielectric functions, `zubko`, `themis` and `g18d` from the $Q_{\rm sca}$ and $\langle\cos\rangle$ columns of their own tables — so all six give a physical albedo. Setting `par%kext_file` replaces them with a tabulated file; the balance above then holds only if that file came from the same model.

Wavelength-dependent extinction is applied as a grey rescaling for each photon

$$s_{\rm ext}(\lambda) = \frac{C_{\rm ext}(\lambda)}{C_{\rm ext}(\lambda_{\rm ref})}, \tag{1}$$

so the grid stores only the reference-wavelength opacity and every optical depth is multiplied by $s_{\rm ext}$ (the same device as the Jonsson 2006 $\tau$ scan). All three ray tracers (Cartesian, clumpy, AMR) are therefore shared with the monochromatic path, and a run with $s_{\rm ext} = 1$ is bit-identical to the mono code.

The source spectrum is either a Planck function at `par%tstar` or a two-column file $\lambda$, $L_\lambda$ (`par%source_spectrum`, or `par%src_spectrum`($i$) for each component). `par%spectrum_type` fixes the column units and the shape-vs-absolute treatment: `'shape'` (default, and always for a Planck source) uses only the shape as the wavelength-sampling distribution while the absolute scale is set by `par%luminosity` / `par%src_lum` (legacy behavior). A physical type is *absolute*: `'per_um'` ($\lambda[\mu\mathrm{m}]$, $L_\lambda\,[\mathrm{erg\,s^{-1}\,\mu m^{-1}}]$), `'per_ang'` ($\lambda[\text{Å}]$, $L_\lambda\,[\mathrm{erg\,s^{-1}\,Å^{-1}}]$), `'per_hz'` ($\nu[\mathrm{Hz}]$, $L_\nu$), or `'per_ev'` ($E[\mathrm{eV}]$, $L_E$); the reader converts to $L_\lambda$ per $\mu$m with the Jacobians $L_\lambda = L_\nu\, c/\lambda^2$ and $L_\lambda = L_E\, hc/\lambda^2$. For an absolute type, when `par%luminosity` / `par%src_lum` is left unset the source luminosity is derived from the file integral over the wavelength grid; when set, the file is rescaled to it. The helper `python/make_source_spectrum.py` writes ready-to-use tables under `data/`: single stellar populations from FSPS (`spec_{young,old}_fsps.dat`) and Starburst99 (`spec_{young,old}_sb99.dat`), and the Mathis et al. (1983) Milky-Way interstellar radiation field (`isrf_mathis.dat`). The `examples/galaxy` models use the FSPS young/old pair and `examples/milkyway` uses the Mathis field.

**External illumination.** SED mode also supports the external geometries (`external_sph`, `external_cyl`, `external_rec`). The external field is an isotropic mean intensity, given either as a shape (`par%ext_spectrum` file or `par%ext_tstar` Planck) scaled by the band mean intensity `par%ext_intensity` $[\mathrm{erg\,s^{-1}\,cm^{-2}\,sr^{-1}}]$, or as an absolute `par%ext_spectrum` file whose $J$ is derived from the file integral when `par%ext_intensity` is unset. In an `par%ext_spectrum` file column 2 is the mean-intensity density $J_\lambda$ in the `par%spectrum_type` units. The spectrum priority is `par%ext_spectrum` file > `par%ext_tstar` Planck > the global `par%source_spectrum`/`par%tstar`. The luminosity entering the grid is

$$L = \pi J A_{\rm surface}, \tag{2}$$

with $A_{\rm surface}$ the illuminated boundary area: $4\pi\,(\texttt{par\%rmax}\cdot\texttt{par\%distance2cm})^2$ for `external_sph`, the six box faces for `external_rec`, and the side plus both end caps for `external_cyl`. This $L$ sets `par%luminosity` for the absolute image normalization.

**Composition (internal source + external field).** An internal source (or sources) and an external isotropic field can illuminate the medium in one run. It is activated when a non-external internal source is present (`par%nsource` ≥

1, par%source_geometry not an external_* value) together with an external field — par%ext_intensity $> 0$ in the monochromatic mode, or par%ext_intensity $> 0$ / par%ext_spectrum / par%ext_tstar in SED mode. par%ext_geometry ('sph'/'cyl'/'rec'; blank auto-selects from par%geometry/par%rmax) fixes the boundary the external field enters through when composing; the external-only run keeps source_geometry='external_*' instead. With the internal total $L_{\rm int}$ (a single par%luminosity, or $\sum_i$ src_lum($i$) for several sources) and the external luminosity

$$L_{\rm ext} = \pi J A_{\rm surface}, \qquad L_{\rm tot} = L_{\rm int} + L_{\rm ext}, \tag{3}$$

where $J$ = par%ext_intensity (or $J$ from par%ext_spectrum in SED mode) and $A_{\rm surface}$ is the same illuminated-boundary area as above, each photon's origin is drawn internal-vs-external in proportion to $L_{\rm int} : L_{\rm ext}$ and every packet carries $L_{\rm tot}/N_{\rm phot}$. par%luminosity is set to $L_{\rm tot}$ for the output normalization, the run reports TOT_LUM = $L_{\rm int} + L_{\rm ext}$, and the composed image equals (internal-only) + (external-only) within Monte-Carlo noise. Composition is incompatible with Stokes polarization and with the $(a, g)/\tau$ scans; pre-existing external-only and internal-only runs are unchanged.

Restrictions: SED mode is incompatible with Stokes polarization and with the $(a, g)/\tau$ single-run scans.

## 5.2 Mean intensity $J_\lambda$ in each cell

| Parameter | Default | Description |
|---|---|---|
| save_jlam | .false. | Tally $J_\lambda$ and write <base>_jlam |

The mean intensity is accumulated with the Lucy (1999) pathlength estimator: each path segment of length $d\ell$ contributes $w L_{\rm pk}\, d\ell$ to its (wavelength-bin, cell) slot, and

$$J_\lambda(\text{cell}) \;=\; \frac{1}{4\pi V_{\rm cell}\, \Delta\lambda\, d_{\rm cm}^2} \sum_{\rm paths} w L_{\rm pk}\, d\ell, \tag{4}$$

where $L_{\rm pk}$ is the packet luminosity [erg/s] and $d_{\rm cm}$ the distance-unit conversion. The forced first-flight is tallied analytically (a zero-variance direct term). A built-in energy check reports the absorbed luminosity from the tally against an independent event counter. Memory is $\approx n_\lambda \times n_{\rm cell} \times 8$ bytes per MPI rank ($n_{\rm cell} = n_x n_y n_z$ on the Cartesian grid, or the number of leaves on the AMR octree; see §5.8).

## 5.3 Mode 1: Lucy (1999) with SEDUST

| Parameter | Default | Description |
|---|---|---|
| use_dustemis | .false. | Compute dust emission (requires par%use_sed) |
| dust_emission_method | 'lucy' | 'lucy' (SEDUST) or 'bw01' (Bjorkman & Wood) |
| dust_model | 'astrodust' | SEDUST grain model: 'astrodust', 'dl07', 'mrn', 'zubko', 'themis', or 'g18d' |
| sed_datadir | blank (auto) | SEDUST data directory, the only path a run names and the data root of the whole build; blank auto-resolves to <MoCafe.x dir>/SEDust/data |

| Parameter | Default | Description |
|---|---|---|
| `sed_dl07_sdindex`, `sed_dl07_uisrf` | 7, 1.0 | 'dl07' only: WD01 size-dist. index and reference $U$ |
| `sed_kext` | blank | Extinction curve to serve; blank takes it from the model's own product |
| `sed_NT, sed_Tlo, sed_Thi` | 200, 2.7, 5000 | Grain temperature grid |
| `dust_niter` | 1 | Lucy iterations (dust self-absorption) |
| `dust_no_photons` | 1e6 | Dust-emission photons per iteration |
| `dust_tol` | 1e-3 | Convergence on the total emitted luminosity |

The `'lucy'` method requires `par%save_jlam`. For each cell the local mean intensity $J_\lambda$ (converted to SI and carried onto the SEDUST wavelength grid bin by bin, so that each transfer bin keeps the energy the packets deposited there) is passed to SEDUST, which returns the emission spectrum from the full grain-size distribution — equilibrium grains plus stochastically heated small grains and PAH features. The *shape* is taken from SEDUST while the bolometric luminosity is set to the locally absorbed power,

$$L_{\rm emit}(\text{cell}) \;=\; L_{\rm abs}(\text{cell}) \;=\; \rho\kappa \sum_{\lambda} s_{\rm ext}(\lambda)\,[1-\varpi(\lambda)] \left( \sum\nolimits_{\rm paths} w L_{\rm pk}\, d\ell \right)_{\lambda,\rm cell}, \tag{5}$$

so total energy is conserved by construction (radiative equilibrium). With `par%dust_niter` $> 1$ the re-emitted photons are transported so that dust self-absorption heats other cells; the iteration converges when the total emitted luminosity stops changing (needed at high infrared optical depth).

**What `par%sed_datadir` names, and what fixes the grid.** The whole of a dust model's optics — its wavelength axis, its $(\lambda, a_{\rm eff})$ cross sections and its size-integrated extinction curve alike — is one file,

`<sed_datadir>/<dust_model>/sedust_<dust_model>.h5`,

so a run names a directory rather than a $Q$ table, a size-distribution file and an extinction table separately. What is not optics comes from beside it: the dielectric functions and PAH cross sections under `dielectric/`, the ZDA config, size distributions and calorimetry under `zubko/`, the DustEM `GRAIN_*.DAT`, wavelength grid and heat capacities under `themis/` and `g18d/`; the HD23 size distribution is analytic (Hensley & Draine 2023, eqs. 17 and 24) and nothing is read for it. That directory is the data root for the whole build, so there is no working directory to change into and nothing else to point at.

That product carries *one* wavelength axis with the Lyman limit marked in it, so the grain model's grid is a view of that axis rather than a choice of file: its non-ionizing part unless `par%lambda_min` asks for shorter wavelengths, and then the whole of it. MoCafe decides that from `par%lambda_min`, there being nothing to set — the grid has to reach at least as far as the transport does, and no further, since the extinction is refused rather than extrapolated outside the grid the model was built on. The two views are 1129 of 1762 wavelengths for astrodust, 1129 of 1823 for dl07 and mrn, which share an axis, 866 of 1201 for zubko, and 750 of 800 for themis and g18d, the cut being an index cut at the last node at or below the Lyman limit, so the grid covers that limit whichever model is named; on

`examples/dustemis/model_compare_astrodust.in` the wider one costs a factor $\simeq 2.4$ in wall time and moves the emitted SED by 0.02% (median over the bins above a tenth of the peak) for as long as nothing is transported below the Lyman limit.

**Emission-solver speed options.** The exact stochastic solve costs $\sim 0.1$ s per cell; two faster paths are available:

| Parameter | Default | Description |
|---|---|---|
| `dust_single_teq` | .false. | One mixture equilibrium $T$ per cell; fastest ($\sim 55\times$); true $T_{\rm eq}$; *no PAH/stochastic features* |
| `dust_fast_table` | .false. | Interpolate emission vs. field intensity $U$ over a fixed reference shape; $\sim 1\%$ SED-shape error; keeps stochastic/PAH |
| `dust_nU` | 40 | $U$-grid size for the table path |

## 5.4 Mode 2: Bjorkman & Wood (2001)

Set `par%dust_emission_method = 'bw01'`. This is an approximate, equilibrium, mixture mean-opacity method with *no iteration*: fixed-energy packets random-walk through scatter and absorb+immediate-reemit events until they escape. Each absorption raises the local temperature (from the running absorbed power) and the packet is given a new, continuous wavelength drawn from the temperature-correction spectrum $\propto \kappa_{\rm abs}\, \partial B_\lambda / \partial T$, so the time-averaged emission equals $\kappa_{\rm abs} B_\lambda(T)$ without iterating. Mode 2 needs only the mixture opacity, not the stochastic solve, and does not reproduce PAH or stochastic-heating features. It writes `<base>_bwdust` (equilibrium $T_{\rm dust}$ and absorbed-power maps); the emergent dust emission is added to the observer `Scattered` SED image.

## 5.5 Multiple stellar populations and galaxy models

| Parameter | Default | Description |
|---|---|---|
| `nsource` | 1 | Number of stellar components ($> 1$ activates multi-source) |
| `src_geometry(i)` | 'point' | Geometry of each source (list below) |
| `src_tstar(i)` | -999 | Planck temperature of each source [K] |
| `src_spectrum(i)` | '' | Spectrum file of each source (overrides src_tstar) |
| `src_lum(i)` | -999 | Luminosity of each source [erg/s] (equal split if unset; derived from the file integral for an absolute spectrum_type file) |
| `src_x/y/z(i)` | 0 | Position of each source ('point') |
| `src_zscale(i)` | -999 | Vertical scale height (exponential/sech/exp_spiral) |
| `src_rscale(i)` | -999 | Radial scale length of the exponential disk |
| `src_sersic_index(i)` | 4 | Sersic index $n$ of a 'sersic' bulge |
| `src_reff(i)` | -999 | Effective radius (sersic) / scale (boxy/bar/xbar) |
| `src_axial_ratio(i)` | 1 | Vertical flattening of a sersic/boxy/bar/xbar bulge |

| Parameter | Default | Description |
|---|---|---|
| src_boxiness(i) | 2 | Boxiness of a boxy/bar/xbar bulge (2 = ellipsoid, >2 = boxy) |

**Source geometries.** point, uniform, gaussian as before; exponential (radially exponential disk $r \sim re^{-r/r_s}$ × vertical exponential); sech (radial disk × sech$^2$); exp_spiral (exponential disk with log-spiral arms); sersic (3-D deprojected Sersic bulge, radius *alias-sampled* from the analytic cumulative); boxy/bar/xbar (generalized-ellipsoid bulges, boxiness src_boxiness).

**Spiral arms and vertical dust profile** (shared by the exp_spiral sources and the cylinder dust density):

| Parameter | Default | Description |
|---|---|---|
| spiral_m | 0 | Number of log-spiral arms (0 = axisymmetric) |
| spiral_pitch | 15 | Pitch angle [deg] |
| spiral_amp | 0.5 | Arm amplitude in $1+\mathrm{amp}\sin(m(\ln r/\tan p-\phi))$ |
| density_zprofile | 'exp' | Vertical dust: 'exp' $e^{-\lvert z\rvert/z_s}$ or 'sech' $\mathrm{sech}^2(z/z_s)$ |

Sources are sampled in proportion to their luminosity, and par%luminosity becomes $\sum_i$src_lum(i). Multiple internal sources work in both the monochromatic and the SED modes. Examples: a disk + bulge SED (examples/galaxy/galaxy_sed.in), a face-on spiral galaxy (spiral_faceon.in), and the Milky-Way all-sky (examples/milkyway/mw_allsky.in).

## 5.6 HEALPix all-sky interior observer (Milky Way)

| Parameter | Default | Description |
|---|---|---|
| allsky | .false. | Observer inside the medium; bin every event by sky direction |
| allsky_x/y/z | 0 | Interior observer position (code units) |
| allsky_nside | 64 | HEALPix resolution (power of 2); $n_{\mathrm{pix}} = 12\,n_{\mathrm{side}}^2$ |

Each scatter, dust-reemission, and direct event peels toward the interior observer and is binned into the **HEALPix** pixel (RING scheme, equal-area pixels) of the sky direction it arrives from, following LaRT's observer_heal implementation. The event-to-observer optical depth is integrated only over the segment to the observer, using the Cartesian or the octree segment ray trace as appropriate — so the all-sky map works on **both the Cartesian grid and the AMR octree**. The output <base>_allsky is a HEALPix ($n_{\mathrm{pix}}, \lambda$) surface-brightness map — the diffuse Galactic light and far-infrared cirrus as seen from inside the disk. It carries the standard HEALPix keywords (PIXTYPE, ORDERING='RING', NSIDE, NPIX, OMEGAPIX) and a PixelDir table of pixel-center unit vectors, so it reads directly with healpy (examples/milkyway/).

## 5.7 Modified Random Walk (high optical depth)

| Parameter | Default | Description |
|---|---|---|
| `use_mrw` | .false. | Diffusion step in scattering-thick cells |
| `mrw_gamma` | 2.0 | Trigger threshold on the nearest-wall scattering optical depth |

In a cell whose nearest-wall *scattering* optical depth exceeds `par%mrw_gamma`, MoCafe replaces the many small steps with a single diffusion step to the inscribed-sphere surface (Min et al. 2009; Robitaille 2010): it samples the total path length inside the sphere from the diffusion first-passage distribution, deposits the absorption-weighted path into the $J_\lambda$ tally, and decays the weight by $\exp(-\alpha_{\rm abs}\, ct)$. The Lucy transport also applies Russian roulette so high-albedo, high-$\tau$ clouds do not diffuse indefinitely. MRW is exact in absorbing dust (where it triggers rarely); its speed-up is large only when individual cells are very thick ($\tau_{\rm cell} \gg 1$) and nearly pure-scattering.

## 5.8 Dust emission on the AMR octree

The entire dust-emission suite — SED transport, the $J_\lambda$ tally, Mode 1 (Lucy, including the single-$T_{\rm eq}$ and table options), Mode 2 (Bjorkman & Wood), and the Modified Random Walk — runs on the AMR octree (`par%grid_type = 'amr'`) exactly as on the Cartesian grid. A "cell" is simply an octree leaf: the radiation field, absorbed power, and dust temperature are tallied per leaf (code-volume and opacity of each leaf), and MRW uses the nearest leaf-face distance for its trigger. Set up the octree as usual (`par%grid_type = 'amr'`, `par%amr_file`, and `par%dust_density_law` for the leaf opacity) and add the emission parameters above. The derived files carry arrays indexed by leaf instead of Cartesian cubes: `<base>_jlam`, `<base>_dustsed`, and `<base>_bwdust` each add a `LeafXYZ` table of leaf-center coordinates so the leaf quantities can be placed in space. Example: `examples/amr_sphere/amr_dustemis.in` (a uniform level-5 octree sphere) reproduces the energy budget and SED of the equivalent Cartesian sphere.

## 5.9 A minimal emission run

```
&parameters
 par%no_photons  = 2.0e6
 par%use_sed     = .true.
 par%save_jlam   = .true.
 par%use_dustemis    = .true.
 par%dust_single_teq = .true.     ! fast equilibrium solve
 par%luminosity  = 3.828e33       ! erg/s
 par%nlambda     = 100
 par%lambda_min  = 0.0912
 par%lambda_max  = 2000.0
 par%dust_model  = 'astrodust'    ! optics and emission from one model
 par%tstar       = 1.0e4
 par%taumax      = 5.0
 par%source_geometry = 'point'
 par%distance_unit   = 'pc'
```

```
 par%rmax = 1.0
 par%nx = 33
 par%ny = 33
 par%nz = 33
 par%distance = 1.0e5
/
```

## 6. Output Products

`par%file_format` is authoritative: if `par%out_file`'s extension disagrees, MoCafe rewrites it (with a warning) so that the main file and all derived files share one container. HDF5 mirrors the FITS HDU layout (each image HDU becomes a group named after its `EXTNAME` with a `data` dataset; FITS keywords become group attributes).

| Suffix | Written when | Contents |
|---|---|---|
| `_obs` | always | Observer images; in SED mode `Scattered`/`Direct` become $(x, y, \lambda)$ cubes plus the wavelength / optics tables |
| `_jlam` | save_jlam | $J_\lambda(\lambda, x, y, z)$, $J_{\rm bol}$, energy-conservation keywords |
| `_dustsed` | use_dustemis ('lucy') | Emergent and intrinsic dust SED, a pixel-resolved `DustEmis_image`$(x, y, \lambda)$, and $T_{\rm dust}/L_{\rm dust}$ maps for each cell |
| `_bwdust` | 'bw01' | Equilibrium $T_{\rm dust}$ and absorbed-power maps |
| `_allsky` | allsky | HEALPix $(n_{\rm pix}, \lambda)$ all-sky surface-brightness map (RING) + `PixelDir` |
| `_obs_tau` | sightline_tau | Sight-line optical-depth map |
| `_stokes` | use_stokes | Stokes $I, Q, U, V$ images |

On the AMR octree the `_jlam`, `_dustsed`, and `_bwdust` images are arrays indexed by leaf rather than Cartesian cubes, and each adds a `LeafXYZ` table of leaf-center coordinates. Read and convert the outputs with `python/mocafe_io.py` (`info`, `peek`, `convert`).

## 7. Worked Examples

| Directory | Shows |
|---|---|
| `examples/point_source`, `uniform_source`, `external_*` | Cartesian grids and source geometries |
| `examples/mono_multi_source` | Monochromatic scattering with two internal point sources (`par%nsource` $> 1$) |
| `examples/compose_int_ext` | Internal source + external field composed in one run (`compose_mono.in`, `compose_sed.in`) |

| Directory | Shows |
|---|---|
| `examples/agtau_list` | $(a, g)$ and $\tau$ single-run scans |
| `examples/clump_sphere`, `amr_sphere`, `amr_tng`, `amr_ramses` | Clumpy and AMR media |
| `examples/dustemis` | Dust emission: Lucy+SEDUST, iteration, single-$T_{\rm eq}$, table, B&W, MRW |
| `examples/sed_external` | External-field SED illumination (Planck shape + `par%ext_intensity`; absolute `per_um` $J$ file) |
| `examples/sed_multi_source` | Multi-source SED with luminosity derived from absolute spectra |
| `examples/amr_sphere` | Dust emission on the AMR octree (`amr_dustemis.in`; matches the Cartesian sphere) |
| `examples/multipop` | Two stellar populations (hot young + cool old) |
| `examples/galaxy` | Galaxy SED (`galaxy_sed.in`: disk + Sersic bulge) and a face-on spiral galaxy (`spiral_faceon.in`) |
| `examples/milkyway` | All-sky map from an interior observer (spiral dust disk) |
| `examples/benchmarks` | Camps (2015) SHG dust-emission benchmark and Mode 1/2 $\tau$ sweep |

The `python/` tools read and plot these outputs: `mocafe_io.py` (reader), `plot_dust_sed.py`, `plot_dust_maps.py`, `plot_jlam.py`, `plot_allsky.py`, `plot_galaxy_sed.py`, and the `quickstart_dustemis.ipynb` notebook.




Figure 1: *Left*: the UV-to-FIR SED of the `examples/galaxy` model (exponential disk + Sersic bulge): the attenuated starlight and the thermal dust emission (PAH bands and a far-infrared peak), with $L_{\rm dust}/L_\star \approx 0.8$. *Right*: the mid-plane dust-emission luminosity of the `spiral_faceon.in` model, showing the $m = 2$ log-spiral arms imposed on the dust density.

## 8. Validation

- **Regression.** With the emission features off, the monochromatic scattering, the $(a, g)/\tau$ scans, and the clumpy-medium outputs are bit-identical to a stand-alone scattering run.

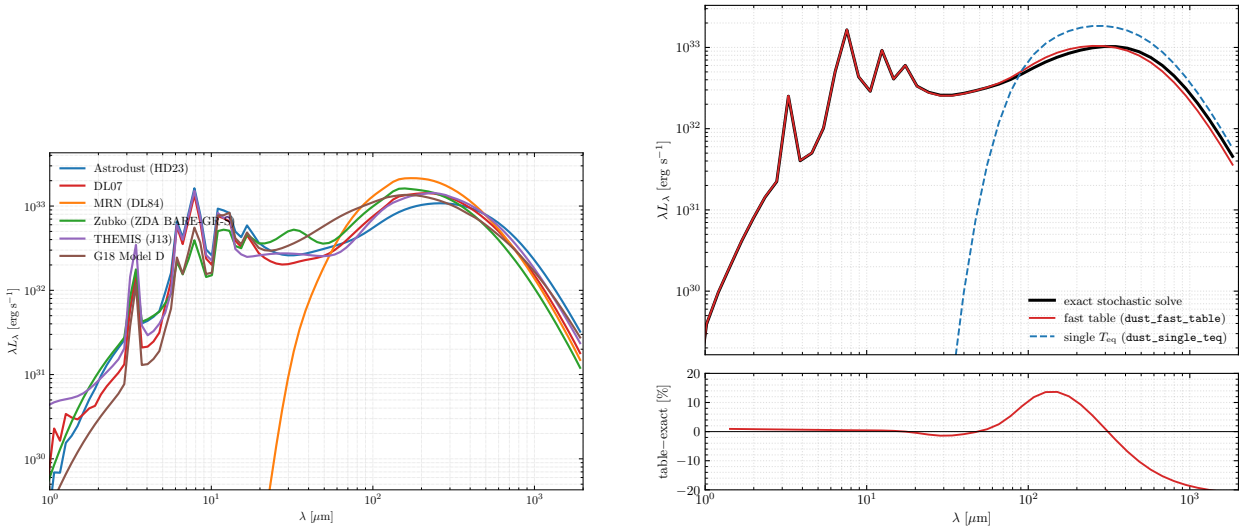



Figure 2: *Left*: the emission spectrum of a single cell heated by the same radiation field, for the six SEDUST grain models (`astrodust`, `dl07`, `mrn`, `zubko`, `themis`, `g18d`); the total power is identical (each is normalized to the absorbed luminosity) while the PAH bands and the far-infrared slope differ by model. *Right*: the `par%dust_fast_table` approximation against the exact stochastic solve — the total luminosity is identical, the PAH/mid-infrared region agrees to $\sim 1\%$, and the largest deviations sit on the far-infrared slope.

- **Dust-emission engine (cross-code).** Run through the SEDUST library with the Zubko et al. (2004) BARE-GR-S model, the emission spectra agree with the Camps et al. (2015) seven-code stochastic-heating benchmark—which includes SKIRT and DIRTY—to a median relative difference of 2.5–4.4% across $U = 10^{-2} \ldots 10^{4}$, inside the code-to-code envelope at every $U$, and the emission peak matches the code median to within one wavelength bin.
- **Energy conservation.** The intrinsic dust SED integrates to the absorbed luminosity to machine precision (the emission is normalized to the locally absorbed power). The Lucy path-length estimator and the Bjorkman & Wood immediate-reemission scheme agree on the absorbed luminosity to $< 2\%$ over the sweep $\tau_V = 0.1 \ldots 20$.
- **AMR vs. Cartesian.** Dust emission on a uniform level-5 octree sphere reproduces the reprocessed fraction (0.863 at $\tau_V = 5$) and energy balance of the equivalent Cartesian sphere; Bjorkman & Wood and MRW on the octree conserve energy identically.

These checks are consolidated in `examples/benchmarks/validate.py`, which re-runs the energy-conservation, two-method, and SHG cross-code tests and reports PASS/FAIL. External full-galaxy transport benchmarks (the TRUST slab, a direct SKIRT/DIRTY image cross-check) need those codes or their published reference solutions and are added as they become available; the emission engine is already cross-checked against SKIRT and DIRTY through the SHG benchmark above.

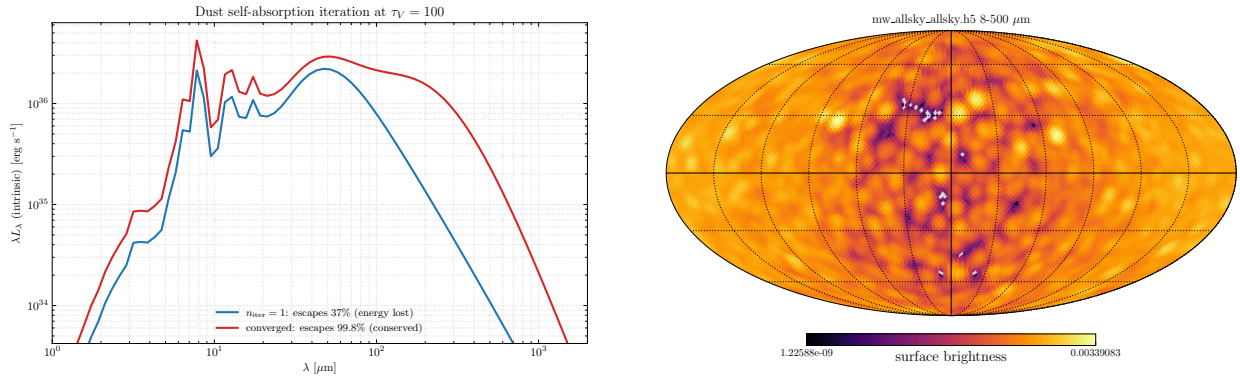



Figure 3: *Left*: convergence of the dust self-absorption iteration (`par%dust_niter`) at $\tau_V = 100$: the emitted luminosity converges geometrically, and only the converged run conserves energy (a single pass loses the reabsorbed dust emission). *Right*: the smoothed 8–500 $\mu$m all-sky map from the interior-observer Milky-Way model (`examples/milkyway`), a HEALPix Mollweide projection showing the Galactic-plane concentration.

# 9. Algorithm Notes

**Packet energy.** Each packet carries a physical luminosity $L_{\rm pk}$ [erg/s]: $L_\star/N_\star$ for stellar packets and $L_{\rm dust}/N_{\rm dust}$ for dust-emission packets, so populations with different packet energies combine correctly in the $J_\lambda$ tally.

**Lucy iteration.** The stellar $J_\lambda$ is tallied once and reused; each iteration transports fresh dust-emission photons (sampled per cell from the emission spectrum), adds their $J_\lambda$ to the stellar field, and recomputes the emission. The total emitted luminosity obeys $L^{(k)} = L^\star_{\rm abs} + f\,L^{(k-1)}$ with $f$ the reabsorbed fraction, a geometric series converging to $L^\star_{\rm abs}/(1-f)$.

**Bjorkman & Wood temperature.** A cell's equilibrium temperature is found by inverting the monotonic $\kappa_B(T) = \int \kappa_{\rm abs}(\lambda) B_\lambda(T)\, d\lambda$ against the running absorbed power per H nucleus.

# References


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- Camps, P. et al. 2015, A&A, 580, A87 — stochastic-heating dust-emission benchmark.
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- Min, M. et al. 2009, A&A, 497, 155; Robitaille, T. P. 2010, A&A, 520, A70 — Modified Random Walk.
- Seon, K.-I. 2010, PKAS, 25, 177 — the $(a, g)$ single-run scan.
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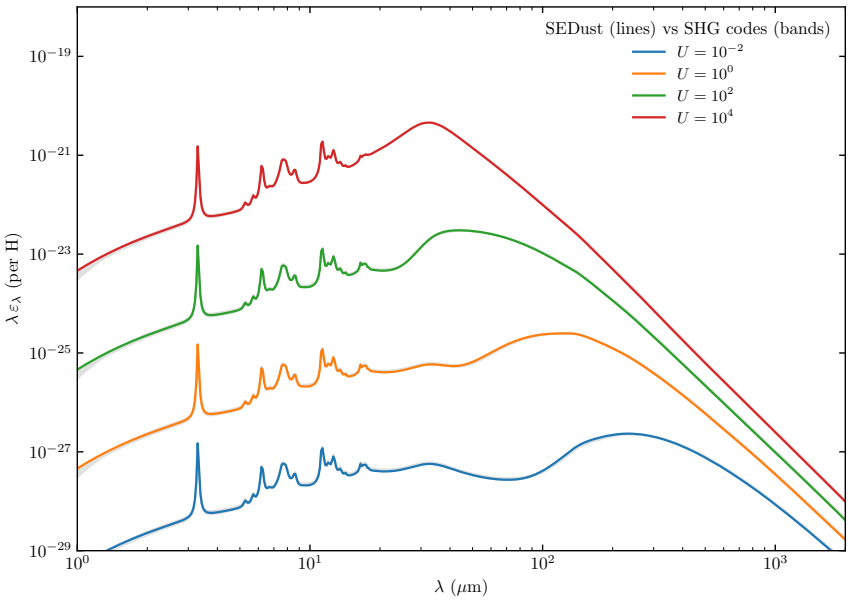



Figure 4: The SEDUST single-cell emission spectrum (zubko BARE-GR-S) against the Camps et al. (2015) seven-code stochastic-heating benchmark at several field strengths $U$; the shaded band is the code-to-code spread, put on our own scale (the published spectra are in the benchmark's units, so the comparison is of shape — the level is set by the absorbed power and is checked separately). MoCafe tracks the benchmark median to a 2.5–4.4% relative difference over $U = 10^{-2} \ldots 10^{4}$.